Memo: Elekeiroz T-2103 butyraldehyde tower: recover the intended 20% capacity increase

2026-11-02 22:16 +03 · DRAFT, not approved

Decision asked for: T-2103 is limited near 9.5 t/h, while the project premise is 11.5 t/h. Build the existing tower, explain why added boil-up does not create useful capacity, and propose a design that reaches the premise without losing isomer separation.

Recommendation: Simulator screening passed; proceed to data and vendor verification (not ready for approval): 1 change to reach 121 % of today's throughput - (1) Replace T-2103 internals with high-capacity trays in same shell.

Result: At 121 %, existing plant: 1 limit over 100 % (highest T-2103 distillation Fair flood fraction vs limit 103 %); with the changes: all modelled process limits within 100 % (highest T-2103 outlet (product leaving the unit) temperature vs design 93 %).

- (1) Replace T-2103 internals with high-capacity trays in same shell - cost class $100k-$1M

Risk: no second-opinion disagreement recorded. Relief findings on changed equipment: 0.

Measure next: Vendor hydraulic rating of HC trays vs RV-1 at 1.9 m, 425 mm spacing; Actual reboiler duty margin at 11.5 t/h.

Full MOC file with Rev A/Rev B drawings and PSSR checklist (6 items) attached. Audit head `58bedf8b4384`.